**COURSE:** MOPP/MOIM
**EXERCISE NAME**: Hand Blow moulding

**EXERCISE NO:** PP-09

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|---|---|---|
| 1 | **Objective** | I) To produce 10 good components using given raw material and blow mould.<br>ii) Calculate production output<br>iii) Enlist the defects during process & remedial measure taken thereof. |
| 2 | **Machine/ eqt./Accessories/ Tools required** | Stopwatch, Vernier caliper, Hand pyrometer with melt and surface probe, bags, weighing balance, plastic resin (grade, MFI, manufacturer, form etc) |
| 3 | **Safety** | 1. Don't touch hot barrel.<br>2. Switch off heater in case of fumes / signs of degradation.<br>3. Wear safety goggles & gloves.<br>4. Follow strictly Do's & don'ts. |
| 4 | **Principle** | Blow moulding is a plastics conversion operation to produce bottles or hollow objects. Molten material is forced through a die in the form of parison which is arrested between two halves of the mould. Parison is blown up by air to get defined part. |
| 5 | **Procedure** | 1. Mould setting is done with proper adjustment and it should be kept airtight when the adjustment is to be performed.<br>2. The timer to keep the heating period at constant depending on the processing range of the raw material controls temperature of the heater.<br>3. Air pressure is set which air compressor for blowing operations provides, which is set through a regulator.<br>4. Material is added in to the barrel and the heater is set to the required temperature.<br>5. As the material is melted the plunger is actuated.<br>6. The long extrudate of the molted material (parison) is brought in between the two mould halves and compressed air is blown into it.<br>7. 7. After providing some cooling time the mould is released for ejection of the component. |
| 6 | **Diagram** | |
| 7 | **Observation** | 1. Air pressure used<br>2. Temperature used |
| 8 | **Result & Conclusion** | 1. Production rate per hour.<br>2. Enlist the actual defects, causes and remedial measures taken thereof during the process. |
| 9 | **Precautions** | 1. Use correct grade of material.<br>2. Ensure the temperature controller is working properly.<br>3. Copper or brass rod is used for cleaning the material.<br>4. Air pressure should be uniform. |

| S. No. | Mould opening time (sec) | Parison time (sec) | Mould closing time (sec) | Blowing time (sec) | Cooling time (sec) | Ejection time (sec) | Total Cycle time (sec) |
|---|---|---|---|---|---|---|---|
| 1 | | | | | | | |
| 2 | | | | | | | |
| 3 | | | | | | | |

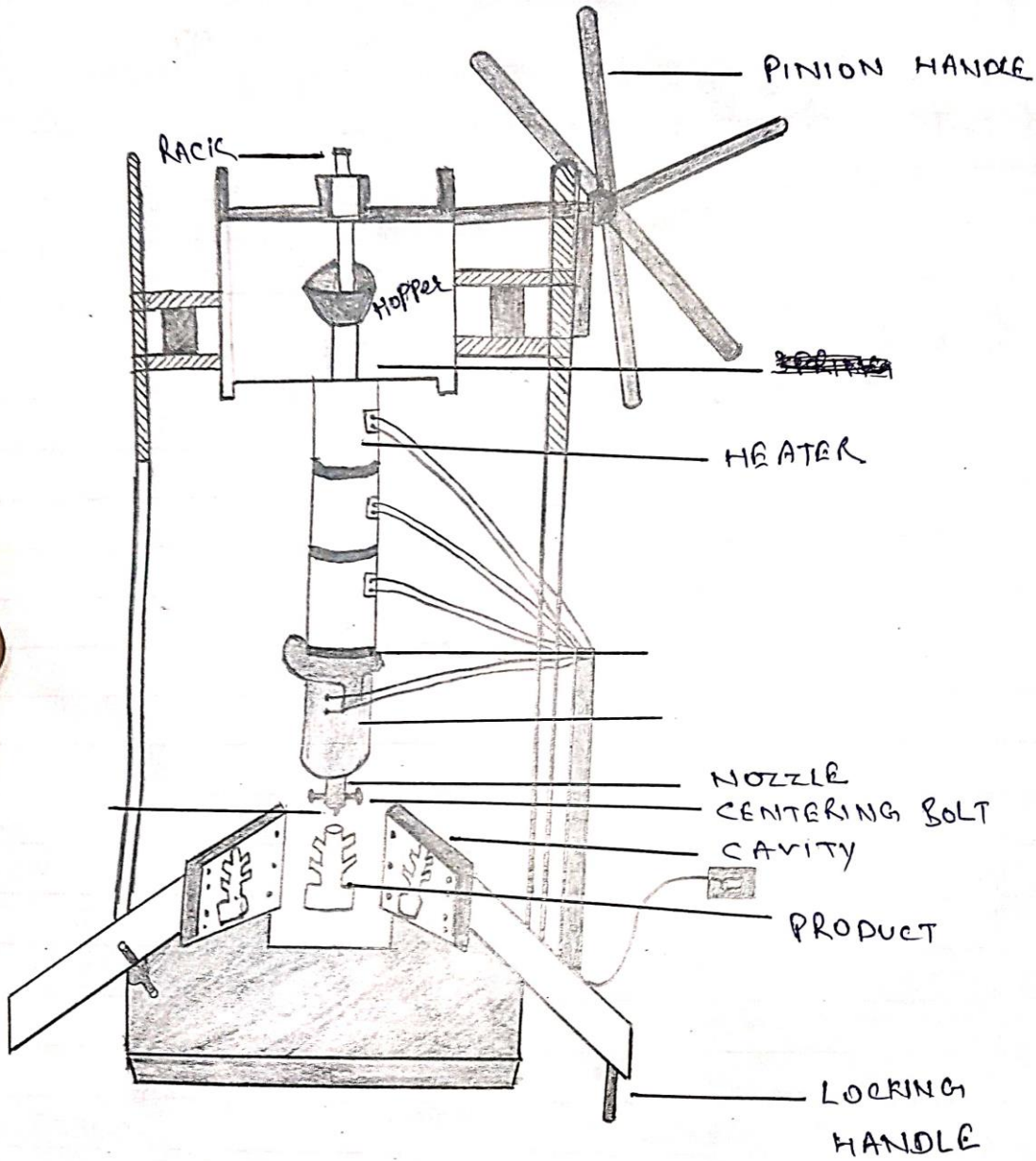
- Production rate per hour = 3600 X No of cavity/ cycle time.
- Calculate blow ratio = Dia. of bottle / Dia. of parison.
- Measure The Die Diameter, Swell Diameter and swell ratio.





**Production :**
**Theoretical values : shift hours / cycle time**
**Actual values : No of articals per hour X 8=**

# HAND INJECTION BLOW MOULDING MACHINE




9/5/24 ** Good.